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HOMEPLOTSAPPSFIGUREFORMATTOOLS

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TitleSubtitleX-LabelY-LabelZ-LabelLegendColorbarGridX-GridY-Grid

FILE

MATLAB Drive > 5G AND 6G LAB

Files

ASSESSMENT.m
ASSESSMENT2.m
EX 1.m

Workspace

Name	Value	Size	Class
avg_delay	309.4296	1×1	double
cloud_co...	1×100 double	1×100	double
cloud_de...	40	1×1	double
cloud_sp...	20	1×1	double
cloud_ta...	100	1×1	double
cloud_to...	1×100 double	1×100	double
cycles	1×100 double	1×100	double
data_rate	1000	1×1	double
decision	1×100 string	1×100	string
edge_co...	1×100 double	1×100	double

ASSESSMENT2.m

```
85 grid on;  
86 xlabel('Task Number');  
87 ylabel('Total Latency (ms)');  
88 title('Edge vs Cloud Processing Latency');  
89 legend('Edge','Cloud');  
90  
91 % Plot 2: Offloading decisions  
92 figure;  
93 bar([edge_tasks cloud_tasks]);  
94 grid on;  
95 xticklabels({'Edge','Cloud'});  
96 ylabel('Number of Tasks');  
97 title('Task Offloading Distribution');  
98  
99 % Plot 3: Selected latency  
100 figure;  
101 plot(1:N,selected_delay,'LineWidth',1.5);  
102 grid on;  
103 xlabel('Task Number');  
104 ylabel('Selected Latency (ms)');
```

Command Window

6	18	1405.14	398.14	CLOUD
7	26	205.21	90.21	CLOUD
8	41	2005.33	540.33	CLOUD
9	46	1405.37	398.37	CLOUD
10	59	2005.47	540.47	CLOUD

Figure 3

Latency After Task Offloading

Selected Latency (ms)

Task Number

Snipping Tool

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EX6.m
EX7.m
EX8.m
ASSESSMENT.m
ASSESSMENT2.m

grid on;
xlabel('Task Number');
ylabel('Total Latency (ms)');
title('Edge vs Cloud Processing Latency');
legend('Edge','Cloud');

% Plot 2: Offloading decisions
figure;
bar([edge_tasks cloud_tasks]);
grid on;
xticklabels({'Edge','Cloud'});
ylabel('Number of Tasks');
title('Task Offloading Distribution');

% Plot 3: Selected latency
figure;
plot(1:N,selected_delay,'LineWidth',1.5);
grid on;
xlabel('Task Number');
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Command Window

6	18	1405.14	390.14	CLOUD
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8	41	2005.33	540.33	CLOUD
9	46	1405.37	390.37	CLOUD
10	59	2005.47	540.47	CLOUD
>>				

Figure 1 x Figure 2 x Figure 3 x

Task Offloading Distribution

Number of Tasks

Edge Cloud

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decision	1x100 string	1x100	string
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Command Window

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7	26	205.21	90.21	CLOUD
8	41	2005.33	540.33	CLOUD
9	46	1405.37	390.37	CLOUD
10	59	2005.47	540.47	CLOUD
>>				

Figure 1
Figure 2
Figure 3

Edge vs Cloud Processing Latency

Task Number	Edge Latency (ms)	Cloud Latency (ms)
1	2000	500
2	2000	500
3	2000	500
4	2000	500
5	2000	500
6	2000	500
7	2000	500
8	2000	500
9	2000	500
10	2000	500
11	2000	500
12	2000	500
13	2000	500
14	2000	500
15	2000	500
16	2000	500
17	2000	500
18	2000	500
19	2000	500
20	2000	500
21	2000	500
22	2000	500
23	2000	500
24	2000	500
25	2000	500
26	2000	500
27	2000	500
28	2000	500
29	2000	500
30	2000	500
31	2000	500
32	2000	500
33	2000	500
34	2000	500
35	2000	500
36	2000	500
37	2000	500
38	2000	500
39	2000	500
40	2000	500
41	2000	500
42	2000	500
43	2000	500
44	2000	500
45	2000	500
46	2000	500
47	2000	500
48	2000	500
49	2000	500
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61	2000	500
62	2000	500
63	2000	500
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65	2000	500
66	2000	500
67	2000	500
68	2000	500
69	2000	500
70	2000	500
71	2000	500
72	2000	500
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75	2000	500
76	2000	500
77	2000	500
78	2000	500
79	2000	500
80	2000	500
81	2000	500
82	2000	500
83	2000	500
84	2000	500
85	2000	500
86	2000	500
87	2000	500
88	2000	500
89	2000	500
90	2000	500
91	2000	500
92	2000	500
93	2000	500
94	2000	500
95	2000	500
96	2000	500
97	2000	500
98	2000	500
99	2000	500
100	2000	500

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Title

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X-Label

Y-Label

Z-Label

Legend

Colorbar

Grid

X-Grid

Y-Grid

Files

ASSESSMENT.m

EX 1.m

Workspace

Name	Value	Size	Class
c	300000000	1×1	double
coverage	100×100 log...	100×100	logical
coverage...	2.7038e+03	1×1	double
d	100×100 do...	100×100	double
D	100×100 do...	100×100	double
fc	2.0000e+09	1×1	double
FSPL	100×100 do...	100×100	double
Gr	5	1×1	double
Gt	20	1×1	double
h	550	1×1	double

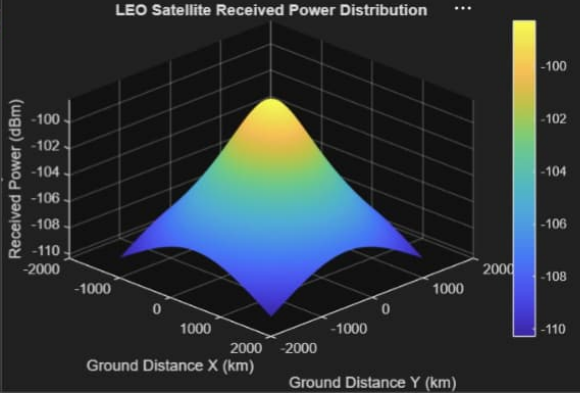
Code Editor

```
68
69 % Received power plot
70 figure;
71 surf(X,Y,Pr,'EdgeColor','none');
72 xlabel('Ground Distance X (km)');
73 ylabel('Ground Distance Y (km)');
74 zlabel('Received Power (dBm)');
75 title('LEO Satellite Received Power Distribution');
76 colorbar;
77 view(45,35);
78
79 % Coverage boundary
80 figure;
81 plot(coverage_radius*cos(theta),coverage_radius*sin(theta),'L
82 hold on;
83 plot(0,0,'o','MarkerSize',10,'LineWidth',2);
84 grid on;
85 axis equal;
86 xlabel('Ground Distance X (km)');
87 ylabel('Ground Distance Y (km)');
```

Figure Window

Figure 1 x Figure 2 x Figure 3 x

LEO Satellite Received Power Distribution



Command Window

```
Carrier Frequency = 2.00 GHz
Transmit Power = 30.00 dBm
Maximum Approximate Coverage Radius = 2703.81 km
Maximum Received Power = -98.28 dBm
Minimum Received Power = -110.28 dBm
>>
```

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